BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) — (GROUP-I) — HSSC(Part-II)-A-2024 (For All Sessions)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8485

Note: NOTE: Write answers to the questions on objective answer sheet provided. Four possible answers A, B, C& D to each question are given. Which answer you consider correct, fill the corresponding circle A, B, C or D given in front of each question with marker or pen ink on the answer sheet provided.

Q1. The geometry of carbonium ion formed in SN^1 mechanism is:

(A) Tetrahedral
(B) Square planar
(C) Triangular planar
(D) Hexagonal

Q2. Ethanol can be converted into enthanoic acid by: [sic]

(A) Hydrogenation
(B) Hydration
(C) Oxidation
(D) Fermentation

Q3. Ketones are prepared by the oxidation of:

(A) Primary alcohol
(B) Secondary alcohol
(C) Tertiary alcohol
(D) Long chain primary alcohol

Q4. Which of the following derivatives cannot be prepared directly from acetic acid?

(A) Acetamide
(B) Acetyl Chloride
(C) Acetic Anhydride
(D) Ethyl Acetate

Q5. Which one of the following elements is NOT present in all proteins?

(A) Carbon
(B) Hydrogen
(C) Nitrogen
(D) Sulphur

Q6. Which is NOT a calcareous material?

(A) Lime
(B) Clay
(C) Marble
(D) Marine shell

Q7. Peroxyacetyl nitrate (PAN) is an irritant to human beings and it affects:

(A) Eyes
(B) Ears
(C) Stomach
(D) Nose

Q8. The main cause of reducing smog is combustion of:

(A) Oils
(B) Coal
(C) Natural gas
(D) Gasoline

Q9. Ionization energy of an atom does not depend on:

(A) Magnitude of nuclear charge
(B) Size of atom
(C) Physical state
(D) Shielding effect

Q10. The oxide of Beryllium is:

(A) Acidic
(B) Basic
(C) Amphoteric
(D) Strongly acidic

Q11. The chief ore of Aluminium is:

(A) Na_3AlF_6
(B) $Al_2O_3.2H_2O$
(C) Al_2O_3
(D) $Al_2O_3.H_2O$

Q12. The oxidation of "NO" in air produces:

(A) N_2O
(B) NO_2
(C) N_2O_3
(D) N_2O_5

Q13. Hydrogen bond is the strongest between the molecules of:

(A) HF
(B) HCl
(C) HBr
(D) HI

Q14. Coordination number of $[Pt(OH)_2(NH_3)_4]SO_4$ is:

(A) II
(B) III
(C) IV
(D) VI

Q15. The state of hybridization of carbon atom in ethane is:

(A) SP^3 (B) sp^2

(C) SP (D) dsp^2

Q16. Preparation of vegetable gnee involves: [sic]

(A) Halogenation (B) Hydrogenation

(C) Hydroxylation (D) Dehydrogenation

Q17. Aromatic hydrocarbons are the derivatives of:

(A) Normal series of paraffins (B) Alkenes

(C) Benzene (D) Cyclohyxane [sic]

Chemistry (Subjective) — (Group-I) — HSSC(Part-II)-A-2024 (For All Sessions)

Time Allowed: 2:40 hours | **Maximum Marks:** 68

Q2. Write short answers of any eight parts from the following: (8x2=16)

i Define hydration energy. How does it vary from top to bottom in I−A and II−A groups?

ii Why do the fluorides show the highest melting and boiling points as compared to other halides?

iii What are the products formed when lithium and sodium nitrates are decomposed?

iv Why is $Mg(OH)_2$ sparingly soluble in water while $Ba(OH)_2$ most soluble?

v What is the difference between paramagnetic and diamagnetic substances? Give brief description.

vi What is meant by co-ordination number and co-ordination sphere?

vii How is ethane prepared from ethyl bromide?

viii How CH_3-CHO react with ethyl magnesium bromide followed by acid hydrolysis?

ix What is difference between Hydrolases and Lysases? [sic]

x Why are lipids important?

xi How temperature and radiation affect the enzymes?

xii What is an acid rain? Give brief description about its impact on our environment.

Q3. Write short answers of any eight parts from the following: (8x2=16)

i Write down any four similarities between oxygen and sulphur.

ii Give two methods of preparation of NO_2.

iii What are Freons and Teflons?

iv Give chemical reaction of NaOH and Cl_2 in hot state (70°C).

v What is modern definition of organic chemistry?

vi Define functional group isomerism with one example.

vii Differentiate between saturated and unsaturated hydrocarbons.

viii What is catalytic oxidation of CH_4 upto the formation of HCHO?

ix The reaction of propene with HBr follow Markownikov's rule. Justify the statement by giving reaction.

x Define environmental pollutant.

xi Write down any two harmful effects of acid rain.

xii How are leather tanneries responsible for pollution of water?

Q4. Write short answers of any six parts from the following: (6x2=12)

i What chemical garden? [as printed]

ii Write down four uses of Sodium Silicate.

iii Why is the aqueous solution of Borax alkaline in nature?

iv Prepare Glyoxal from benzene.

v How are ethers prepared by Williamson's synthesis?

vi Why is phenol acidic in nature?

vii Give the reactions of Formaldehyde with: (i) HCN ii) NH_2-OH.

viii How is acetamide prepared from acetic acid?

ix Give the two reactions in which H-atom of carboxylic acid is involved.

Q. SECTION-II

Note: Attempt any three questions. Each question carries equal marks: (8x3=24)

Q5. (a) Write a note on oxides as a periodic relationship in compounds. (4)

(b) Why Lithium shows peculiar behaviour? Give its any seven differences from other alkali metals. (4)

Q6. (a) Give eight uses of Nobel gases. (b) How is urea fertilizer is prepared in Pakistan? Describe the process in detail. (4+4)

Q7. (a) Discuss structure of methane on the basis of hybridization. (4)

(b) How Propyl Magnesium Bromide reacts with following? (i) CH_3COCH_3 (ii) CO_2. (2+2)

Q8. (a) How does ethyne react with (i) Hydrogen (ii) Halogen acid (iii) alkaline $KMnO_4$ (iv) 10% H_2SO_4 + $HgSO_4$. (1x4)

(b) How sodium bisulphite reacts with (i) Formaldehyde (ii) Acetone Also write the general mechanism. (4)

Q9. (a) Define Friedel-Crafts acylation alongwith its example and mechanism. (4)

(b) How can you prepare the following from Phenol? (i) Benzene (ii) Cyclohexanol (iii) 2, 4, 6-tribromopheol [sic] (iv) [illegible in source scan] phenoxide. (1x4)

GROUP – II

Chemistry (Objective) — (GROUP-II) — HSSC(Part-II)-A-2024 (For All Sessions)

Time Allowed: 20 Minutes	**Maximum Marks:** 17	**Paper Code:** 8484

Note: NOTE: Write answers to the questions on objective answer sheet provided. Four possible answers A, B, C& D to each question are given. Which answer you consider correct, fill the corresponding circle A, B, C or D given in front of each question with marker or pen ink on the answer sheet provided.

Q1. Which one of the following is NOT present in all proteins?

(A) Carbon
(B) Hydrogen
(C) Nitrogen
(D) Sulphur

Q2. For which crop, Ammonium Nitrate fertilizer is NOT used?

(A) Cotton
(B) Wheat
(C) Sugarcane
(D) Paddy rice

Q3. PAN is an irritant to human beings and it affects:

(A) Eyes
(B) Ears
(C) Stomach
(D) Nose

Q4. In purification of potable water, the coagulant used is:

(A) Nickel Sulphate
(B) Copper Sulphate
(C) Barium Sulphate
(D) Aluminium Sulphate

Q5. The correct statement is:

(A) $Na^+ < Na$
(B) $Na^+ > Na$
(C) $C\bar{l} < Cl$
(D) $C\bar{l} = Cl$

Q6. The oxide of Be is:

(A) Acidic
(B) Basic
(C) Amphoteric
(D) Neutral

Q7. The aluminium oxide is:

(A) Acidic
(B) Amphoteric
(C) Basic
(D) Neutral

Q8. The laughing gas is chemically:

(A) NO
(B) NO_2
(C) N_2O_4
(D) N_2O

Q9. Which halogen occurs naturally in a positive oxidation state?

(A) I_2
(B) F_2
(C) Br_2
(D) Cl_2

Q10. The strength of binding energy of transition elements depends upon:

(A) No. of electron pairs
(B) No. of protons
(C) No. of neutrons
(D) No. of unpaired electrons

Q11. The chemist who synthesized urea from ammonium cyanate was:

(A) Berzelius
(B) Kolbe
(C) Wholer
(D) Lavoisier

Q12. Synthetic rubber is made by polymerization of:

(A) Chloroform
(B) Acetylene
(C) Divinyl acetylene
(D) Chloroprene

Q13. During nitration of benzene, the active nitrating agent is:

(A) NO_3
(B) NO_2^+
(C) NO_2
(D) HNO_3

Q14. For which mechanisms, the first step involved is the same:

(A) E_1 and E_2
(B) E_2 and SN_2
(C) SN_1 and E_2
(D) E_1 and SN_1

Q15. Ethanol can be converted into ethanoic acid by:

(A) Hydrogenation
(B) Hydration
(C) Oxidation
(D) Fermentation

Q16. Ketones are prepared by the oxidation of:

(A) Primary alcohol (B) Secondary alcohol

(C) Tertiary alcohol (D) Dimethyl ether

Q17. Which of the following is not a fatty acid?

(A) Propanoic acid (B) Acetic acid

(C) Phthalic acid (D) Butanoic acid

Chemistry (Subjective) — (GROUP-II) — HSSC(Part-II)-A/2024 (For All Sessions)

Time Allowed: 2:40 hours | **Maximum Marks:** 68

Q2. Write short answers of any eight parts from the following: (8x2=16)

i What are polymeric hydrides? Give example.

ii What are the oxidation states of group VII A, and VIII A?

iii What happens when Lithium Hydroxide is heated to red hot?

iv Why is the aqueous solution of Na_2CO_3 alkaline in nature?

v Why metal chelates are more stable than metal complexes?

vi How Potassium Permanganate oxidizes oxalic acid?

vii Give IUPAC names of following compounds: (i). $(CH_3)_2CHBr$ (ii). CH_2Cl_2

viii How is carboxylic acid prepared from carbon dioxide and Grignard's reagent?

ix What are the reasons for rancidity of oils and fats?

x Write down any two functions of DNA.

xi What is the repeating unit of polystyrene and Teflon?

xii What are fertilizers? Why do we need them?

Q3. Write short answers of any eight parts from the following: (8x2=16)

i Give ring test for the confirmation of the presence of nitrate ion in solution.

ii P_2O_5 is a powerful dehydrating agent. Prove it by giving example.

iii Why HF is weaker acid than HCL? Justify. [sic]

iv What are disproportionation reactions? Give example.

v Write short note on steam cracking.

vi What are heterocyclic compounds? Give two examples.

vii Convert Methyl Magnesium Bromide into methane.

viii State Markownikov's rule, give an example.

ix Write a short note on acidity of ethyne.

x How is oil spillage affecting the marine life?

xi Are detergents are threat to aquatic life? Justify. [sic]

xii Mention any two conditions which are required for the formation of smog.

Q4. Write short answers of any six parts from the following: (6x2=12)

i What is water glass and how it is prepared?

ii Why aluminium is used in making petrol and milk storage tanks?

iii How is boric acid dehydrated stepwise when heated strongly?

iv Give two objections to Kekule's formula of benzene.

v Why are lower alcohols readily soluble in water?

vi How would you prepare Bakelite from phenol?

vii Prepare acetaldehyde and acetone by dry distillation method.

viii Give two reactions of amino acids.

ix Differentiate between complete reduction and partial reduction of acetic acid.

Q. SECTION-II

Note: Attempt any three questions. Each question carries equal marks: (8x3=24)

Q5. (a) Write down two similarities and two dissimilarities of hydrogen with Group IA elements. (04)

(b) Give the four points in which Beryllium differs from the other members of its own family. (04)

Q6. (a) Explain relative reactivities of the halogens as oxidizing agent. (04)

(b) Explain the process of setting of cement & give reactions taking place in first 24 hours and 1-7 days. (04)

Q7. (a) Discuss the structure of ethyne on the basis of hybridization. (04)

(b) Discuss two main factors which govern the reactivity of alkyl halides. (2+2=4)

Q8. (a) Explain free radical mechanism for the reaction of chlorine with methane in the presence of sunlight. (04)

(b) How HCN reacts with (i) Formaldehyde (ii) Acetone? Also write down the mechanism. (2+2=4)

Q9. (a) Predict the major products of bromination of the following compounds: (a) Toluene (b) Nitrobenzene (c) Phenol (d) Benzaldehyde [sic] (04)

(b) What are the alcohols? Write the reaction of alcohols which show: (i) Oxidation (ii) Dehydration (iii) Substitution (04)